

Multimodal Alignment and Fusion: A Survey

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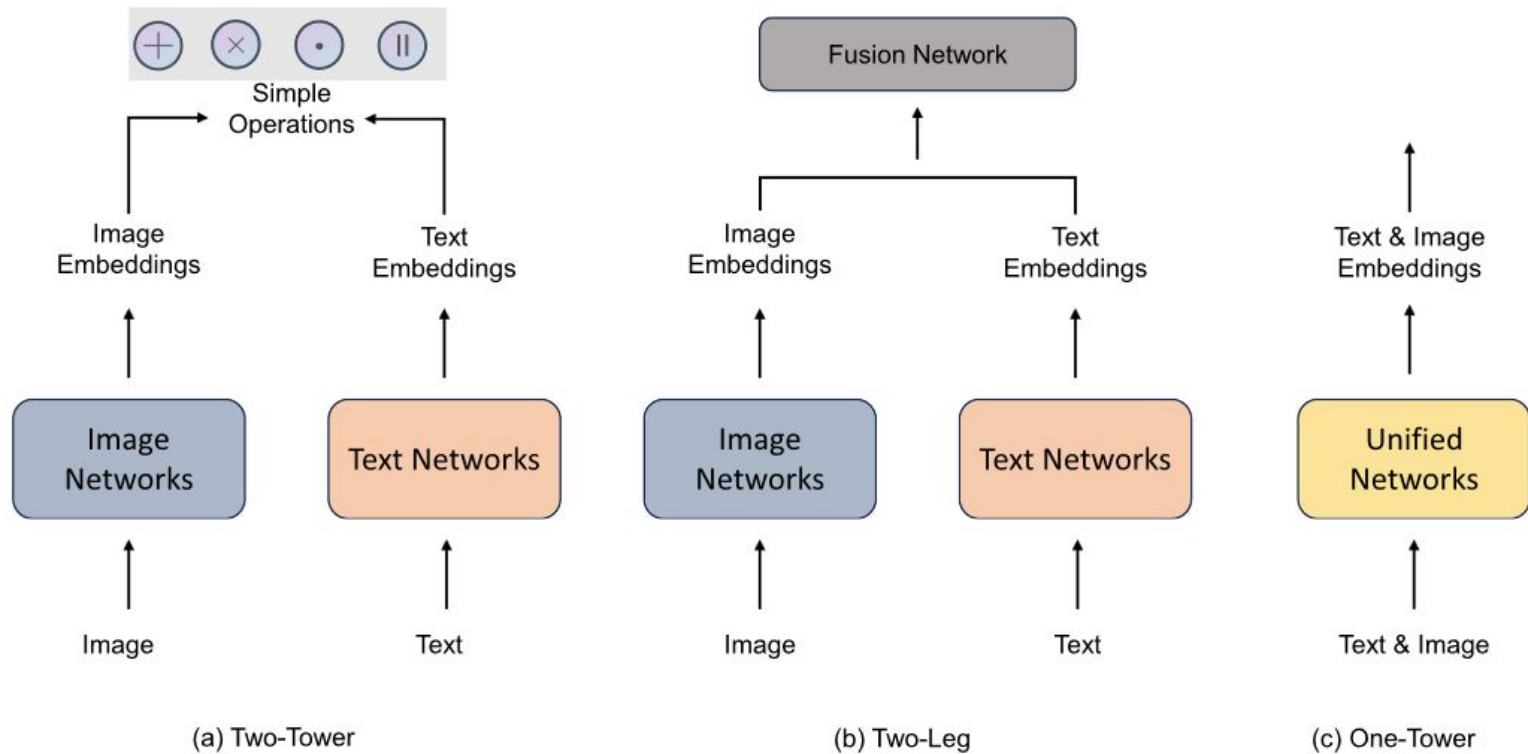


Fig. 1 Overview of multimodal model architectures: (a) Two-Tower (Radford et al., 2021; Jia et al., 2021; Cui et al., 2024; Vasilakis and Bittner, 2024; Xu et al., 2023; Su et al., 2023; Liu et al., 2023a; Chen et al., 2024; Fei et al., 2022; Wen et al., 2024; Tu et al., 2022; Yuan et al., 2021): processes images and text separately, combining embeddings through simple operations (add, multiple, dot product and concatenate); (b) Two-Leg (Allaire and Willcox, 2012; Badrinarayanan and Kendall, 2015; Danapal et al., 2020; Guo and Zhang, 2023; Jaiswal et al., 2015;

Li et al., 2020b; Li and Wu, 2018; Mai et al., 2019; Makris et al., 2011; Missaoui et al., 2010; Rövid and Remeli, 2019; Steinbaeck et al., 2018; Uezato et al., 2020; Wei et al., 2021; Kim et al., 2021): combines separate image and text embeddings using a Fusion Network; (c) One-Tower (Bao et al., 2022; Li et al., 2022, 2023b; Dai et al., 2023; Chen and Xu, 2023; Zhu et al., 2023; Wang and Jiahui, 2022; Wang et al., 2024a; Bai et al., 2023; Alayrac et al., 2022): utilizes a unified network to jointly embed image and text inputs

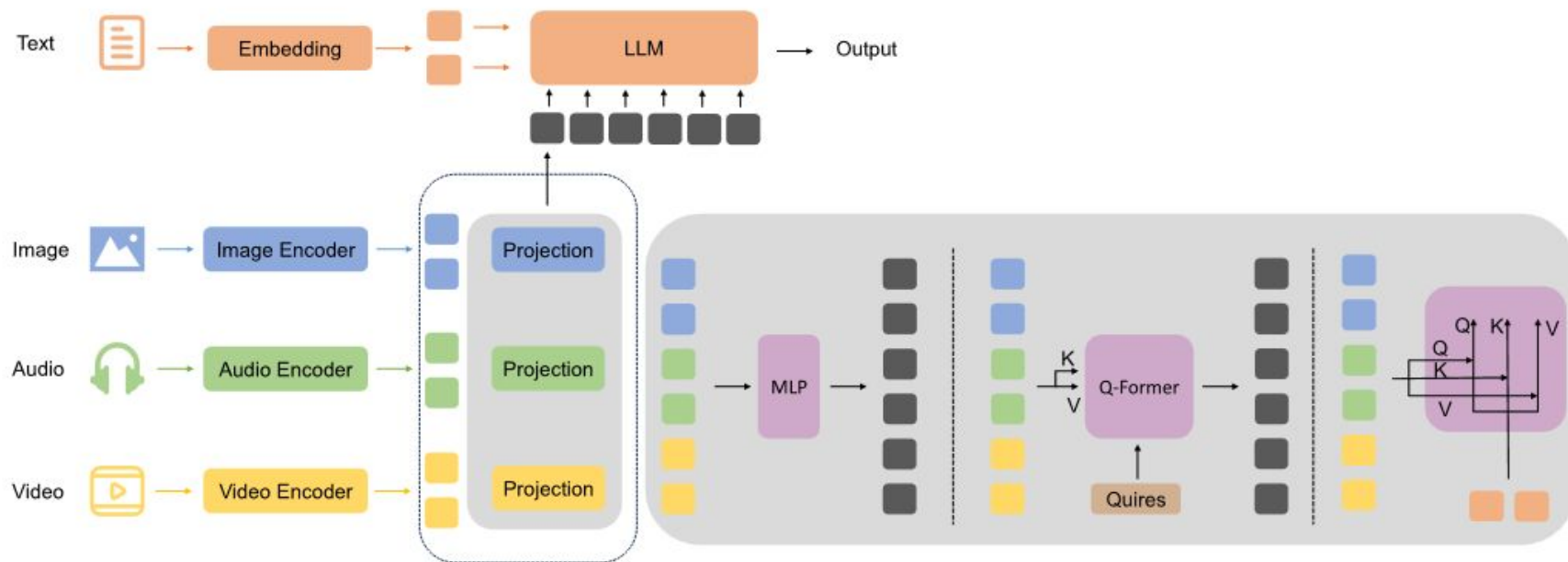
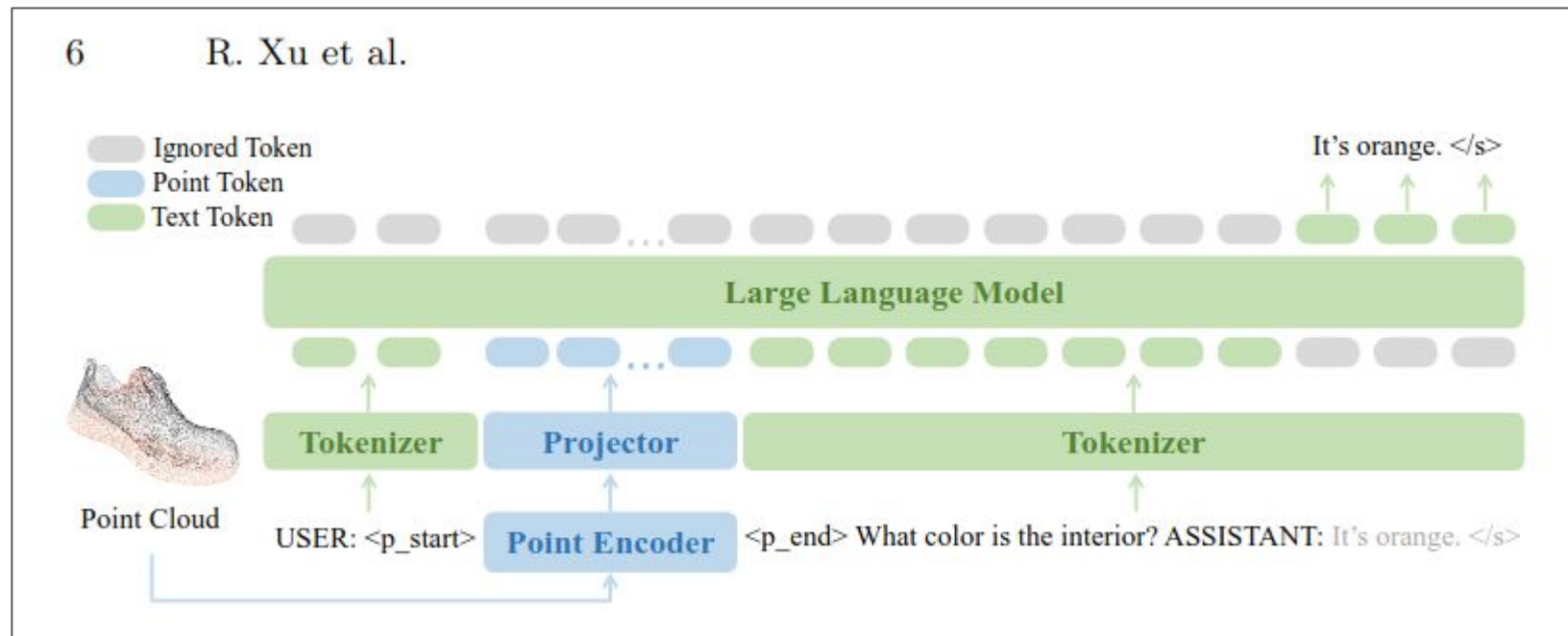


Fig. 7 This pipeline demonstrates multimodal fusion using a large language model (LLM). Text inputs are embedded and processed by the LLM, while image, audio, and video inputs are encoded, projected into a shared embedding space, and passed through modules such as an MLP

and Q-Former. The Q-Former uses attention mechanisms (queries, keys, and values) to align multimodal features before generating a final output through the LLM

PointLlm



Key challenge: aligning point clouds tokens to text token space.

Subset experiments (5K samples / 2 epochs) – BLEU-4/METEOR/ROUGE-L $\times 100$, CIDEr as-is. Deltas vs baseline (r0: BLEU 15.14, CIDEr 0.127):

Run	Change	BLEU-4	CIDEr	Δ BLEU-4	Δ CIDEr	Verdict
r0	baseline: prepend, Q32, 2L GELU	15.14	0.127	–	–	–
r1	queries 64	15.14	0.107	–0.01	–0.02	neutral
r2	queries 128	13.53	0.056	–1.61	–0.07	hurts (dilution)
r3	MLP 3L	15.65	0.119	+0.50	–0.01	~tie
r6	SwiGLU FFN	14.70	0.141	–0.45	+0.01	~tie (slot METEOR)
r16	prepend + proj-lr 1e-2	14.73	0.079	–0.41	–0.05	hurts (overshoot)
r18	prepend + proj-lr 2e-2	15.52	0.099	+0.37	–0.03	hurts
r4	gated x-attn (every 4th)	15.81	0.165	+0.67	+0.04	best subset
r7c	x-attn + gate warmup	16.13	0.135	+0.99	+0.01	best subset BLEU
r11	x-attn + proj-lr 1e-2 + SwiGLU	15.95	0.159	+0.80	+0.03	best balanced
r14	x-attn + gate-lr 1e-3	15.48	0.145	+0.34	+0.02	best val CE (1.997)
r8	x-attn only every 14th (~19M)	15.30	0.094	+0.15	–0.03	insufficient capacity

Full-data runs (660K / 3 epochs):

Run	Arch	BLEU-4	METEOR	ROUGE-L	CIDEr
reference	prepend, Q32, GELU (current)	21.80	46.74	49.45	0.877
r11_fulldata	x-attn + proj-lr + SwiGLU	17.33	41.09	44.05	0.314

Fig A.2 - PointBERT embedding space (umap, WITH projection)

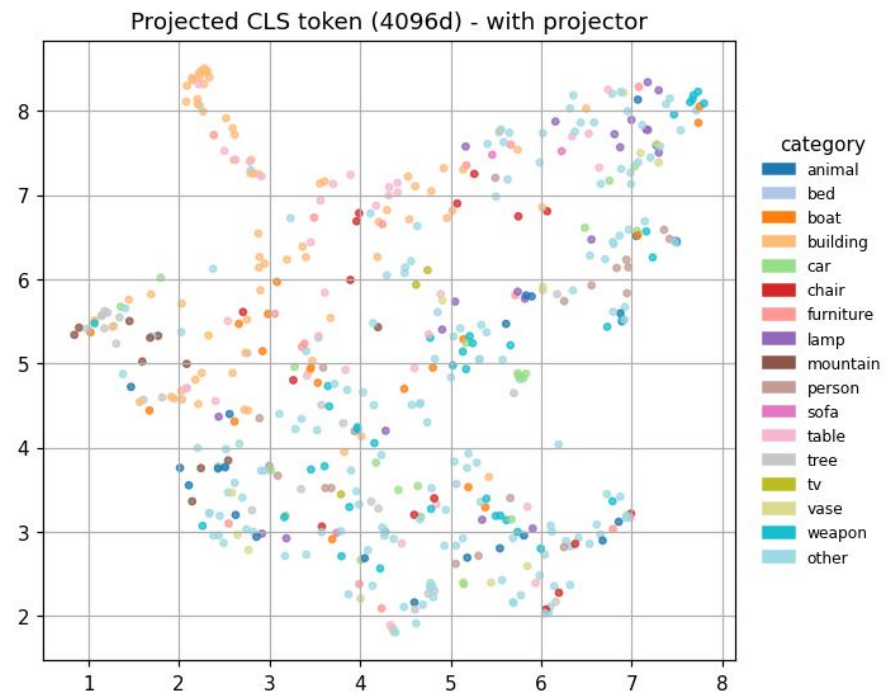
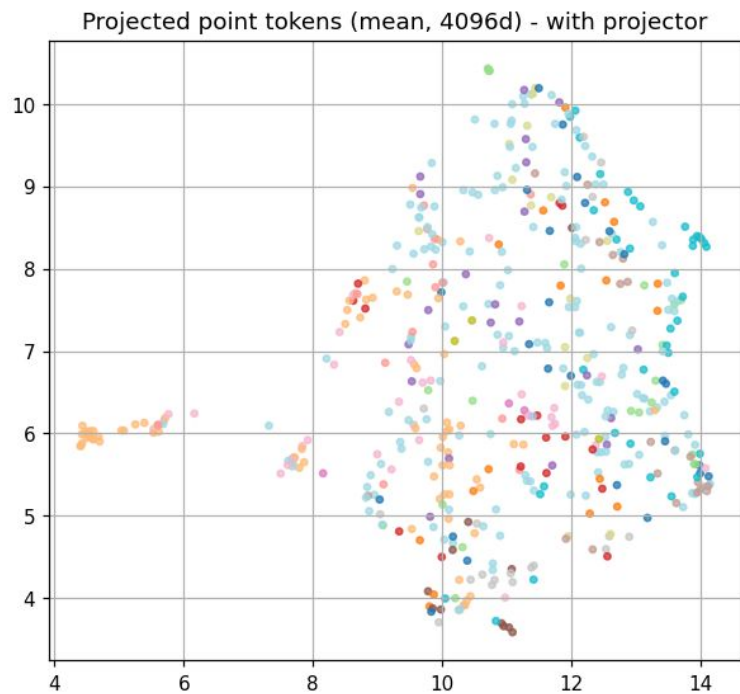


Fig C - point vs text in shared space (umap, 4096d)

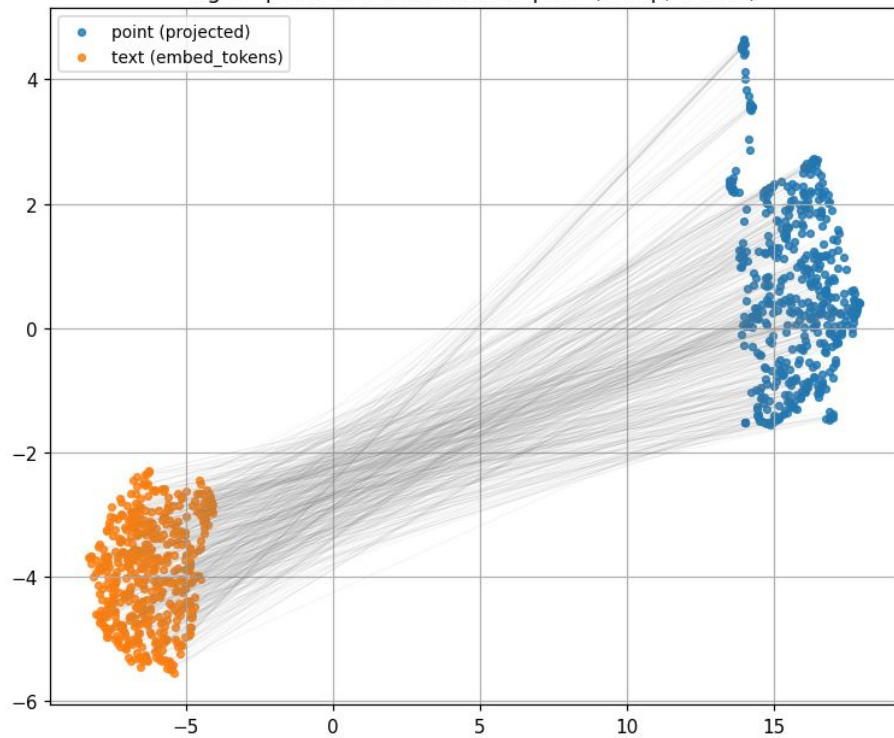


Fig D - combined space colored by category (umap)

